



GUJARAT POLICE INNOVATION CHALLENGE 2026 · CCTV INTEGRATION HACKATHON

# One platform over Gujarat's CCTV — without asking 26 departments to replace theirs

A federated camera registry and GIS, live analytics at the edge, watchlist alerts and cross-camera vehicle tracking, running today on the 30-camera government feed.

## THE PROBLEM

# Many systems, no shared view, no shared memory

**26**

government departments, each with its own CCTV and VMS

**~80,000**

cameras to reach statewide, analog and IP, many vendors

**~1,000 km**

between the furthest sites, on links that are not uniform

- Vendors, AMC periods and storage differ; retention runs from 7 days to 15 or more.
- No department can see another's cameras, or check what a camera sees against the records that would make it actionable.
- The brief: one searchable platform, analysed and alerting in real time, that grows toward 80,000 cameras without a redesign.

### The constraint that shapes everything

Departments keep running their own systems. The first department to join gets value on day one, and the twenty-sixth changes nothing for the first.

# A hybrid: Model 1 foundation, Model 3 integration, Model 2 console

1

## Registry & GIS

Adopted in full

Mandatory, and correct: nothing else works without knowing which cameras exist, where, and who owns them. Useful on day one with no analytics at all.

3

## VMS federation

The integration

One adapter per vendor, normalising to one camera model. Vendor differences stop at the adapter; a department joins without touching anything downstream.

2

## Unified viewing

As the surface

One console for operators — but it talks to the platform, never directly to 26 VMS, so a new vendor is never a change to the screen people use.

4

## Central VMS

Rejected

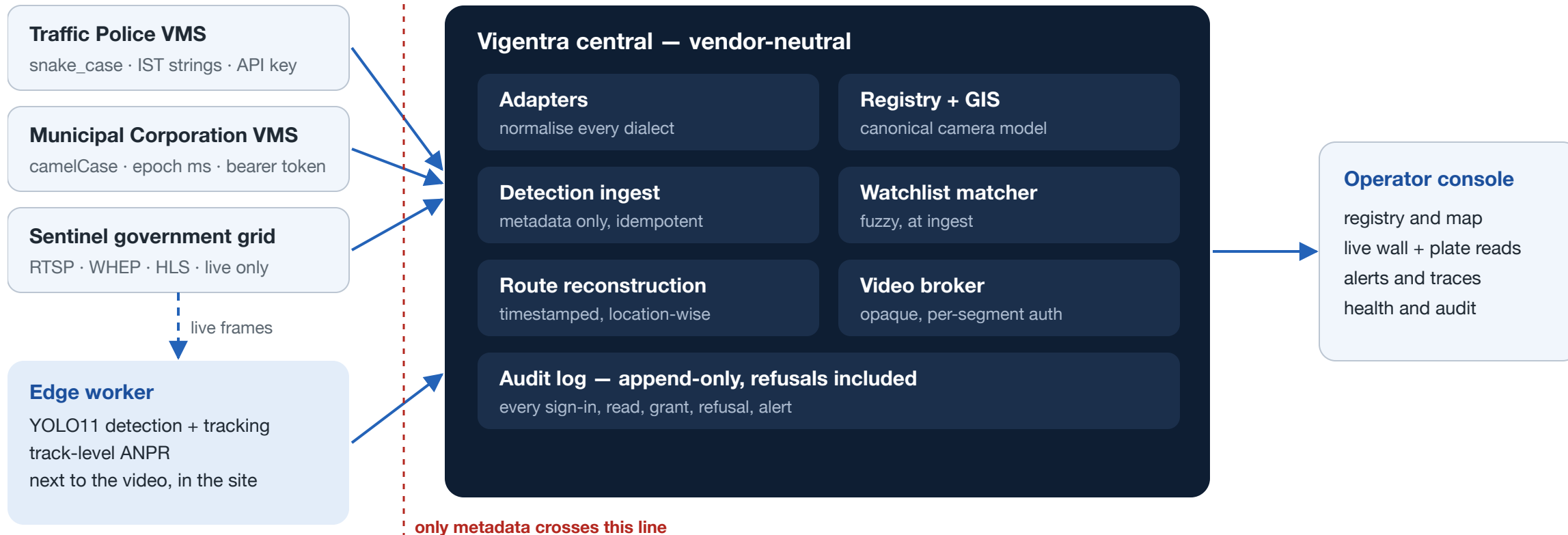
Streaming 80,000 cameras centrally is ~160 Gbps and ~1.7 PB a day, and puts every camera credential in one place. Inference goes to the video instead.

Video is pulled on demand, per authorised session. Analytics run next to the cameras and send only metadata.

## SOLUTION OVERVIEW

# Frames stay where they are; metadata travels

### DEPARTMENT SYSTEMS



# One adapter per vendor; above it, every camera looks the same

|                    | TRAFFIC POLICE         | MUNICIPAL CORPORATION | SENTINEL GRID                              |
|--------------------|------------------------|-----------------------|--------------------------------------------|
| Fields             | flat snake_case        | nested camelCase      | minimal catalogue                          |
| Time               | IST wall-clock strings | epoch milliseconds    | absent — PTS from the stream               |
| Auth               | API key header         | bearer token          | one session per account                    |
| Register / archive | yes / yes              | yes / yes             | none / none — read-only, live-only in code |

### Analog and legacy

An analog camera is reached through the DVR or encoder that digitises it, which exposes RTSP or ONVIF like any IP source.

### Outages stay local

One department down leaves the rest working; its cameras go offline with a stated reason, and every failed poll is audited.

### Liveness follows the video

If the grid's web gateway refuses but its video server answers, cameras stay online — this recovered 72 of 180 lost minutes on 14 Sep.

# ANPR that reads a vehicle, not a frame

## 1 · Detect & track

YOLO11 + ByteTrack  
give each vehicle a  
stable id



## 2 · Find the plate

only on vehicles  $\geq 96$   
px wide, where a plate  
can be read



## 3 · Bank the crops

every crop stamped  
with frame, PTS and  
quality



## 4 · Restore

rectify, register, fuse,  
deglare, deblur



## 5 · Read

CRNN readers on every  
variant; a second  
reader fills gaps



## 6 · Vote

Indian plate grammar +  
consensus across the  
track

## Grammar, not guesswork

A character's position says letter or digit, so  
**GJ05JV34S6** can only be GJ05JV3456.  
Impossible states and districts are dropped at  
the edge.

## One vehicle, one reading

Emitted once per vehicle with the number of  
frames that agreed. Unconfirmed readings are  
shown as unconfirmed, never as fact.

## Object detection

Person, car, motorcycle, bus, truck, auto-  
rickshaw, bicycle — with a frame-quality router  
that skips frames too degraded to mean  
anything.

Skipping plate search on vehicles too small to read gave the same 36 readings on the labelled Delhi clip in 586 s instead of 878 s.

# Known at the moment of the read; traced across cameras



## Matched at ingest

- Every plate reading is checked against the active watchlist as it arrives — stolen, wanted, missing, blacklisted.
- Fuzzy on purpose: a known OCR confusion ( $B \leftrightarrow 8$ ,  $O \leftrightarrow 0$ ) costs 0.35, an unrelated character 1.0.
- Alerts say whether the match was exact, and near matches are shown, not hidden.



## The route, from the plate

- Given a registration: every permitted sighting in time order, with distance, elapsed time and speed per leg, on the map.
- Repeated reads at one camera collapse into one pass; impossible legs are flagged, never dropped.
- Every trace needs a written reason, recorded against the account.

This is the hackathon-day test case: a designated vehicle's complete route and timestamped, location-wise movement history.

# Nobody sees more as cameras are added

1

## Brokered, opaque video

Short-lived, watermarked sessions; every segment is re-authorised, so revoking a grant stops a playing stream.

2

## Scope in three dimensions

Department, city and zone, checked per request, plus each camera's own access policy.

3

## Least privilege

13 roles, 22 permissions. The account that ingests plates cannot read the watchlist back.

4

## Withheld, not refused

Without plate permission an officer still gets the alert, with the plate withheld.

5

## Retention on read

Plates keep a shorter clock, applied at read time — a failed purge cannot extend it.

6

## A complete audit trail

Every sign-in, read, grant, refusal and machine-raised alert, each committed on its own.

# Live, measured, and honest about the limits

30/30

grid cameras onboarded,  
online

96,691

vehicle and object detections,  
20 cameras

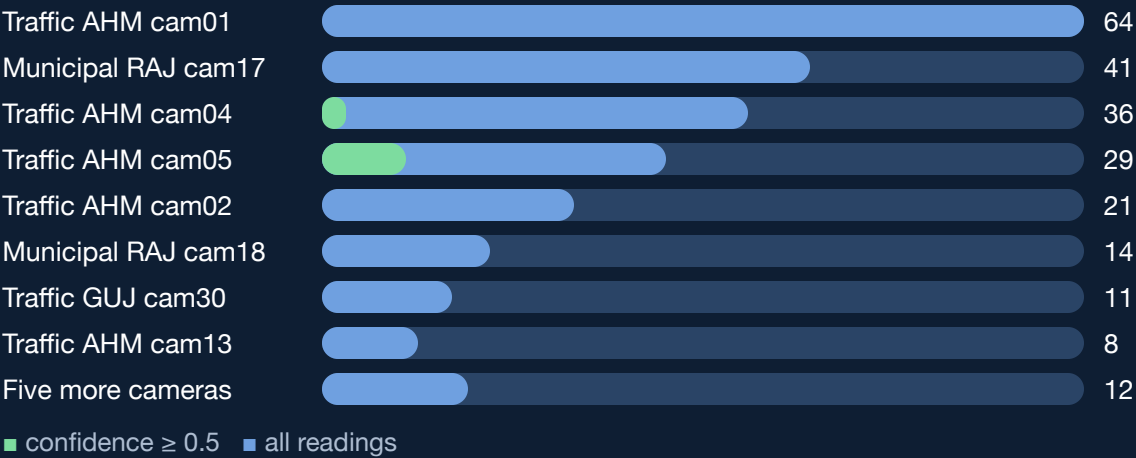
236

plate readings from 12  
cameras, 9 at confidence  $\geq$   
0.5

295

automated tests passing

## Plate readings per camera, 14 Sep



Floors, not capacity: the grid refused access in recurring 15–20 minute windows that day. Most readings are low-confidence because wide junction views make plates only a few pixels wide.

## PLAN FOR SCALE

# To ~80,000 cameras without moving the video

**Streaming everything centrally**

**160 Gbps**

sustained, and ~1.7 PB of video per day

**Vigentra: metadata only**

**0.6–1 Gbps**

statewide, 40–250× less; departments keep their footage

**~3,000 GPUs**

ANPR only where the optics allow,  
classified at onboarding (estimate)

**Hot · warm · cold**

30 days, 1 year, 7 years; plates on  
a shorter clock

**RPO 5 min · RTO 1 h**

synchronous standby, point-in-  
time recovery, DR site

**Watched at scale**

per-camera health history, liveness  
from the video, audited

Named and planned: trigram prefilter for fuzzy plate search, bloom filter for the watchlist, PostGIS for corridor queries.

# Each phase is useful on its own

## Phase 1 · ~50 cameras

one district, all four models exercised — this hackathon

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## Phase 2 · ~2,000

3 districts, 2 departments — genuinely different VMS vendors

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## Phase 3 · ~15,000

all districts, Home Department — statewide load, DR failover tested

›

## Phase 4 · ~80,000

all 26 departments — steady state

## Model 1 goes first, in full, at every phase

An accurate statewide camera inventory with coverage-gap analysis is valuable on the day it exists — and nobody has one today. Analytics follow the inventory.

# Built, running, and verifiable in the code

## Hybrid architecture

Models 1, 3 and 2 together, each where it fits

## Cross-camera tracking

routes from plate reads, impossible legs flagged

## Edge processing

inference beside the video, on CPU or GPU

## Bandwidth optimisation

metadata only; video on demand; sampling dial

## Security and audit

brokered sessions, fine roles, append-only trail

## Automated alerts

watchlist matched at ingest, exact and near

## Health monitoring

per-source and per-camera, with history

## Operational dashboards

registry, map, live wall, alerts, reports

## Integration-ready APIs

documented REST API and adapter contract

# What we believe — and what the grid taught us

## Start from the people who use it

A control-room officer needs one screen; twenty-six departments should not have to change theirs to give it to them. Federation, not replacement, is what makes that possible on day one.

## Never show more certainty than we have

An unconfirmed reading is shown as unconfirmed, a plate too small to read is marked unreadable instead of guessed, and every figure in this deck was measured or is labelled as an estimate.

## Privacy is the design, not a feature

Footage is asked for, never ambient. Every look is brokered and audited — refusals included — because the thing that scales worst in surveillance is how many people can see everything.

## The hard part was the plumbing

On the government feed the model was rarely the problem: refused sessions, dropped streams and a shared account were. The platform now keeps cameras online while video flows and restarts its own readers.

# Everything here runs — check it

## Source and design

- [github.com/rewardix-ai/vigentra](https://github.com/rewardix-ai/vigentra)
- High-level design: [docs/hld.md](#)
- Scalability plan: [docs/scalability.md](#)

## Demonstrations

- Own-feed and government-feed recordings: links in the submission
- Output report: plates and vehicles with timestamps